

Confidence-Weighted Multi-Scale Rolling Forecasting of Urban Taxi Demand with GraphSAGE-Based Spatial Refinement

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MSc by Research Thesis

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Submitted in partial fulfilment of the requirements
for the degree of Master of Science by Research

04 2026

Abstract

Accurate urban taxi demand forecasting is essential for intelligent transportation management, fleet dispatching, and congestion mitigation. However, this task remains challenging because demand patterns exhibit strong temporal non-stationarity, multi-scale periodicity, and complex spatial dependency across city regions. To address these issues, this study proposes a two-stage framework for hourly taxi demand prediction over New York City Taxi Zones, integrating multi-scale temporal modeling, uncertainty estimation, and graph-based spatial refinement. In the first stage, a persistent per-zone forecasting model is developed to capture temporal dynamics at four different scales: hourly, daily, weekly, and monthly. The architecture combines GRU, LSTM, and Transformer components to learn short-term fluctuations and long-range temporal regularities within a unified multi-branch design. Monte Carlo Dropout is further employed to estimate predictive uncertainty for each zone. In the second stage, the base predictions are treated as priors and refined through a GraphSAGE-based residual learning module constructed on an origin-destination flow graph. To improve robustness, a confidence-weighting mechanism is introduced by combining historical sufficiency, model stability, and neighborhood consistency, allowing unreliable zones to contribute less during graph refinement. The overall system is evaluated under a rolling forecasting protocol that simulates real-world hourly prediction updates. Experimental results indicate that the proposed graph refinement stage reduces prediction error relative to the temporal baseline and is especially beneficial for zones with sparse history or high predictive variance. In addition, the proposed method improves spatial coherence among neighboring zones and provides a more stable forecasting process under dynamic urban conditions. Overall, the framework demonstrates that combining multi-scale sequence learning with confidence-aware graph neural refinement offers an effective, robust, and extensible solution for spatiotemporal urban taxi demand forecasting problems.

Acknowledgements

I would like to thank ...

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Chapter 1

Introduction

1.1 Background

Write your introduction here.

1.2 Research Problem

State the research problem clearly.

1.3 Contributions

The main contributions of this thesis are:

- Contribution 1
- Contribution 2
- Contribution 3

Chapter 2

Literature Review

2.1 Related Work

You can cite papers like this `example_ref`.

Chapter 3

Methodology

3.1 Problem Formulation

An example equation:

$$y = f(x) \tag{3.1}$$

3.2 Model Architecture

You can insert figures like this:

Chapter 4

Implementation

4.1 System Setup

Describe your implementation details here.

4.2 Code Example

Example Python code:

```
1      import torch
2      import torch.nn as nn
3
4      class MyModel(nn.Module):
5      def __init__(self):
6      super().__init__()
7      self.linear = nn.Linear(10, 1)
8
9      def forward(self, x):
10     return self.linear(x)
11
```

Listing 4.1: Example Python code

Chapter 5

Experiments

5.1 Experimental Setup

Describe dataset, metrics, baselines, etc.

5.2 Results

Example table:

Table 5.1: Comparison of model performance

Model	MAE	RMSE
Baseline	12.34	18.56
Proposed Method	10.21	15.43

Chapter 6

Discussion

Discuss strengths, limitations, and interpretation of results.

Chapter 7

Conclusion

Summarise the thesis and suggest future work.

Appendix A

Additional Materials

Additional experiments, derivations, or implementation details.